Integer equations

L3-NT2 Level 3 Senior · Number theory

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November 2026

Solve equations in integers by factorising, using parity and remainders to rule out cases, bounding, and finding every solution of ax + by = c.
By the end you can rearrange an equation into a product equal to a constant and list every solution; use parity and small moduli to prove an equation has no solutions or to cut down cases; bound variables to finitely many cases and solve linear equations ax + by = c completely.
Prerequisites: Factors, primes and divisibility (L2-NT1), Factorising, substitution and systems (L2-AL1), Modular arithmetic (L3-NT1). **Contests:** UKMT SMC, Senior Kangaroo, Maclaurin, AMC 12, AIME. **Time:** about 90 minutes.

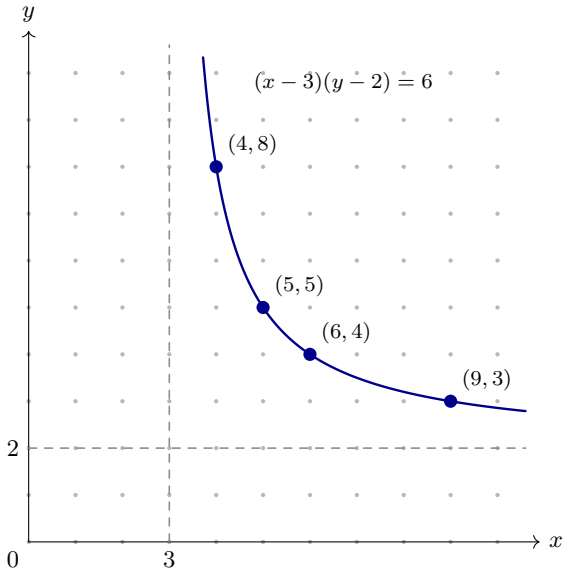
1 Lesson

Find all pairs of positive integers (x, y) with $xy = 2x + 3y$.

Over the real numbers this has infinitely many solutions, since we may choose any $x \neq 3$ and put $y = 2x/(x-3)$. Over the integers, move everything to one side to get $xy - 2x - 3y = 0$, and note that $(x-3)(y-2) = xy - 2x - 3y + 6$. So the equation says

$$(x-3)(y-2) = 6.$$

Two integers multiply to 6, so each is a divisor of 6, and there are only finitely many of those. Since $x, y \geq 1$ we have $x - 3 \geq -2$ and $y - 2 \geq -1$, so a negative factorisation of 6 would need a factor below its allowed minimum, and hence both factors are positive. The pairs $1 \cdot 6$, $2 \cdot 3$, $3 \cdot 2$, $6 \cdot 1$ give $(x, y) = (4, 8), (5, 5), (6, 4), (9, 3)$, the complete list. The curve has infinitely many points, but only four lattice points.



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At this level there are three main methods, namely factorising (a product equal to a constant), remainders (to rule out or restrict cases) and bounding (to reduce to finitely many cases). Linear equations $ax + by = c$ have a complete theory of their own.

1.1 Factorising to make a product equal a constant

The principle.
If $AB = N$, where A and B are integers and $N \neq 0$ is fixed, then A is a divisor of N (positive or negative) and $B = N/A$. So there are finitely many possibilities, and you can list them.

The difficulty is in getting the equation into that shape. Three standard moves cover most cases.

- *Completing the product.* Since $(x+b)(y+a) = xy + ax + by + ab$, the equation $xy + ax + by = c$ becomes $(x+b)(y+a) = c + ab$. If the xy term has a coefficient, multiply through by that coefficient first.
- *Difference of two squares.* $x^2 - y^2 = N$ becomes $(x-y)(x+y) = N$.
- *Completing the square first.* For example $x^2 + 6x = y^2$ becomes $(x+3)^2 - y^2 = 9$, a difference of two squares.

A difference of squares gives extra information. The numbers $x - y$ and $x + y$ differ by $2y$, so they have the same parity.

Theorem.
An integer N is a difference of two squares of integers if and only if N is odd or a multiple of 4.

Proof. Every square is $\equiv 0$ or $1 \pmod 4$, so a difference of two squares is $\equiv 0, 1$ or $3 \pmod 4$, never 2. Conversely, $2k+1 = (k+1)^2 - k^2$ and $4k = (k+1)^2 - (k-1)^2$. ■

When you list factor pairs $N = de$ with $d = x - y$ and $e = x + y$, keep only the pairs where d and e have the same parity. Then $x = (d+e)/2$ and $y = (e-d)/2$.

Example 1.
Find all positive integers n for which $n^2 + 20n + 11$ is a perfect square.

Solution.
Trying $n = 1, 2, 3, 4$ gives $32, 55, 80, 107$, with no sign of when we could stop. So set $n^2 + 20n + 11 = m^2$ with $m \geq 0$ and complete the square to get $(n+10)^2 - 89 = m^2$, that is,

$$(n + 10 - m)(n + 10 + m) = 89.$$

The second factor is positive, so the first is too, and the second is the larger. Since 89 is prime, the only option is $n + 10 + m = 89$ and $n + 10 - m = 1$. Adding, $2(n+10) = 90$, so $n = 35$ and $m = 44$. We check that $35^2 + 700 + 11 = 1936 = 44^2$. The only answer is $n = 35$.
There is a second method, which still works when factorising does not. We have $n^2 + 20n + 11 < (n+10)^2$ for every n, and $(n+9)^2 = n^2 + 18n + 81$ is less than $n^2 + 20n + 11$ exactly when $2n > 70$. So for $n > 35$ the number lies between two consecutive squares, equal to neither, and cannot be a square. That leaves only $n \leq 35$ to check. This squeeze between consecutive squares is a standard bounding method.

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1.2 Remainders, parity and other moduli

If an equation has a solution in integers, then the equation has a solution modulo every n. So if one modulus makes the two sides disagree whatever the variables are, there are no solutions at all. This is the standard way to prove impossibility, and the same idea also cuts down cases when solutions exist.

Squares and cubes modulo small numbers.

- Modulo 3 and modulo 4, every square is 0 or 1.
- Modulo 8, every square is 0, 1 or 4, and every odd square is 1.
- Modulo 7 and modulo 9, every cube is 0, 1 or -1.

Each is checked by running through the residues. The odd-square fact is worth remembering with its proof, namely $(2k+1)^2 = 4k(k+1)+1$, and $k(k+1)$ is even, so 8 divides $4k(k+1)$.

Example 2.
Show that the equation $x^2 - 3y^2 = 2026$ has no solutions in integers.

Solution.
We look for a modulus in which the equation is impossible. The coefficient 3 suggests modulo 3 first, where the equation becomes $x^2 \equiv 2026 \equiv 1 \pmod 3$, which is possible.
Next try modulo 4, where $-3 \equiv 1$. We need $x^2 + y^2 \equiv 2026 \equiv 2 \pmod 4$. Squares are 0 or 1 modulo 4, so this forces $x^2 \equiv y^2 \equiv 1$, which means x and y are both odd. This is not yet a contradiction, but modulo 8 can build on this.
With x and y odd, $x^2 \equiv y^2 \equiv 1 \pmod 8$, so $x^2 - 3y^2 \equiv 1 - 3 = -2 \equiv 6 \pmod 8$. But $2026 = 8 \cdot 253 + 2 \equiv 2 \pmod 8$. This is a contradiction, so there are no solutions.
When a modulus gives no contradiction, use what the modulus does tell you to choose the next one. For squares, modulo 4 followed by modulo 8 is a common pair.

1.3 Linear equations $ax + by = c$

Theorem.
Let a and b be positive integers with $d = \gcd(a,b)$. The equation $ax + by = c$ has integer solutions if and only if $d \mid c$. If (x_0, y_0) is one solution, then all the solutions are

$$x = x_0 + \frac{b}{d}\,t, \qquad y = y_0 - \frac{a}{d}\,t, \qquad t \in \mathbb{Z}.$$

Proof. If there is a solution, then d divides $ax + by = c$. Conversely, suppose $d \mid c$. Dividing everything by d, we may assume $\gcd(a,b) = 1$. By the theorem on inverses in L3-NT1, the congruence $ax \equiv c \pmod b$ has a solution x_0. Then $c - ax_0$ is a multiple of b, say by_0, and (x_0, y_0) is a solution. Now let (x,y) be any solution of the original equation. Subtracting $ax_0 + by_0 = c$ gives $a(x - x_0) = b(y_0 - y)$, and dividing by d gives $\frac{a}{d}(x - x_0) = \frac{b}{d}(y_0 - y)$. Since $\frac{a}{d}$ and $\frac{b}{d}$ are coprime, $\frac{b}{d}$ divides $x - x_0$. Writing $x - x_0 = \frac{b}{d}t$ forces $y_0 - y = \frac{a}{d}t$. Every integer t works. ■

So consecutive solutions differ by b/d in x and a/d in y, in opposite directions, and counting solutions with $x, y \geq 0$ means finding the range of t. To find the first solution, reduce modulo one of the coefficients.

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Example 3.
Pens cost 17p and pencils cost 11p. In how many ways can I spend exactly £10, buying at least one of each?

Solution.
We need positive integers x, y with $17x + 11y = 1000$. Guessing is slow with numbers this size, so we find one solution by working modulo 11. We have $17x \equiv 6x$ and $1000 = 90 \cdot 11 + 10$, so $6x \equiv 10 \pmod{11}$. Since $6 \cdot 2 = 12 \equiv 1$, the inverse of 6 is 2, and $x \equiv 20 \equiv 9 \pmod{11}$. Try $x = 9$. Then $1000 - 153 = 847 = 11 \cdot 77$, so $(9, 77)$ is a solution.
Since $\gcd(17, 11) = 1$, the general solution is $x = 9 + 11t$, $y = 77 - 17t$. We need $x \geq 1$, so $t \geq 0$, and $y \geq 1$, so $17t \leq 76$, that is, $t \leq 4$. Hence $t = 0, 1, 2, 3, 4$, giving $(9, 77)$, $(20, 60)$, $(31, 43)$, $(42, 26)$ and $(53, 9)$. There are 5 ways.
The step sizes have a concrete meaning. Here 11 pens and 17 pencils both cost 187p, so swapping one bundle for the other keeps the total fixed.

1.4 Bounding

When one side grows faster than the other, or the variables can be put in order, you get an inequality that leaves finitely many cases. Two habits help. If the equation is symmetric, assume an order such as $a \leq b \leq c$ and permute at the end. Then replace every variable by the largest (or smallest) one to obtain a bound.

Example 4.
Find all triples of positive integers (a, b, c) with $a \leq b \leq c$ and $abc = 4(a + b + c)$.

Solution.
Three unknowns and one equation, so we must bound something. Since c is the largest, $a + b + c \leq 3c$, so $abc \leq 12c$, and dividing by c gives $ab \leq 12$. In particular $a^2 \leq ab \leq 12$, so $a \leq 3$. Each of the three cases becomes a two-variable product equation.
If $a = 1$, then $bc = 4 + 4b + 4c$, so $(b - 4)(c - 4) = 20$. Neither factor is zero. If both were negative, then $b, c \leq 3$ and the product would be at most $3 \cdot 3 = 9$, so both are positive. With $b \leq c$ we need $(b - 4)^2 \leq 20$, so $b - 4 \in \{1, 2, 4\}$, giving $(b, c) = (5, 24), (6, 14), (8, 9)$.
If $a = 2$, then $2bc = 8 + 4b + 4c$, so $bc - 2b - 2c = 4$ and $(b - 2)(c - 2) = 8$. Here $b \geq 2$, and $b = 2$ would give $0 = 8$, so both factors are positive. With $b \leq c$ we need $b - 2 \in \{1, 2\}$, giving $(b, c) = (3, 10)$ and $(4, 6)$.
If $a = 3$, then $3bc = 12 + 4b + 4c$, and this case needs one extra step. Multiply by 3 to complete the product, giving $9bc - 12b - 12c = 36$, so $(3b - 4)(3c - 4) = 52$. With $3 \leq b \leq c$ the first factor satisfies $5 \leq 3b - 4 \leq \sqrt{52} < 8$, and the only value of the form $3b - 4$ in that range is 5, which does not divide 52. No solutions.
So the solutions are $(1, 5, 24)$, $(1, 6, 14)$, $(1, 8, 9)$, $(2, 3, 10)$ and $(2, 4, 6)$, and each checks, for instance $2 \cdot 4 \cdot 6 = 48 = 4 \cdot 12$. A bound reducing to a few cases, followed by factorising in each case, is typical of the harder SMC and AIME questions.

Common traps.

- Forgetting negative factor pairs when the variables are any integers, or not ruling them out properly when they are positive.
- Ignoring parity in a difference of squares, and listing factor pairs that give non-integer

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x and y.

- Dividing by something that might be zero, such as $x - 3$ in the opening problem.
- Assuming $a \leq b \leq c$ when the equation is not symmetric, or forgetting to count the permutations when the question asks for ordered triples.
- Stopping once you have found some solutions. "Find all" means proving there are no others, and checking each candidate.
- In $ax + by = c$ with $\gcd(a, b) = d > 1$, stepping by b instead of b/d.

1.5 How to spot this in a contest

The main signal is "integers" or "positive integers" beside an equation with more unknowns than equations. Then read its shape.

- A product xy together with linear terms, so complete the product.
- Squares and a constant, or "is a perfect square" suggest a difference of squares, or a squeeze between consecutive squares.
- "Show that there are no solutions", so try small moduli, usually 3, 4, 8, 9 and 7.
- A symmetric equation in several variables, so order them and bound.
- Coins, stamps or prices give a linear equation, solved with one particular solution plus the general step.

In the AMC, "how many ordered pairs" usually means counting divisors of a constant, doubled if negative factors are allowed.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ How many ordered pairs of positive integers (x, y) satisfy $xy = 4x + 6y$?

Problem 2 ⋆⋆ How many pairs of positive integers (x, y) satisfy $x^2 = y^2 + 6y + 129$?

Problem 3 ⋆⋆ A post office sells only 7p and 12p stamps. In how many different ways can a parcel be given exactly £5 of postage? (Two ways are the same if they use the same number of each stamp.)

Problem 4 ⋆⋆⋆ Find all positive integers n for which $n! + 5n + 13$ is a perfect square.

Problem 5 ⋆⋆⋆ Find all pairs of positive integers (x, y) such that $x^3 - y^3 = 721$.

Problem 6 ⋆⋆⋆⋆ How many ordered triples of positive integers (a, b, c) satisfy $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{7}{10}$?

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3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 12A 2015 Problem 10** ⋆
   Make the left-hand side into a product.
2. **AMC 10A 2006 Problem 22** ⋆⋆
   Write a debt as an integer combination of 300 and 210, and ask which totals are possible.
3. **AIME II 2012 Problem 1** ⋆⋆
   Find one solution, write the general one, then count the positive ones.
4. **AMC 12B 2020 Problem 8** ⋆⋆
   Complete the square in y and bound the other term.
5. **AIME II 2000 Problem 2** ⋆⋆⋆
   Difference of squares, and mind the parity of the two factors and their signs.
6. **UKMT SMC 2026 Question 23** ⋆⋆⋆
   Two Pythagoras equations share a height, so eliminate the height and factorise.
7. **AIME 1987 Problem 5** ⋆⋆⋆
   Collect the terms so that one side factorises, then use the prime factorisation of the constant.
8. **AMC 10B 2023 Problem 14** ⋆⋆⋆
   If both variables are large the right-hand side wins, so bound one of them.
9. **BMO1 2009/10 Problem 1** ⋆⋆⋆
   Rewrite the first equation so that a sum of squares equals a small constant.
10. **BMO1 2012/13 Problem 4** ⋆⋆⋆
    Combine the two expressions to eliminate n, then factorise.
11. **BMO1 2016/17 Problem 3** ⋆⋆⋆
    Complete the square on both sides, clearing fractions first.
12. **BMO1 2018/19 Problem 3** ⋆⋆⋆⋆
    Write both products using completed squares and factorise the difference.

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4 Hints

1. Move everything to one side and add a constant so that the left-hand side factorises.
2. Complete the square on the right-hand side, then factorise a difference of two squares. What do you know about the parities of the two factors?
3. Solve $7x + 12y = 500$ in non-negative integers. Working modulo 7 tells you what y must be.
4. For large n, $n!$ is divisible by many small numbers. Which modulus makes both $n!$ and $5n$ vanish?
5. Factorise the left-hand side. Then compare the sizes of the two factors to show that $x - y$ must be small.
6. Count the triples with $a \le b \le c$ first, using $\frac{3}{a} \ge \frac{7}{10}$ to bound a, then a similar bound for b in each case. Count orderings at the end.

5 Solutions

Problem 1

How many ordered pairs of positive integers (x, y) satisfy $xy = 4x + 6y$?

The equation is $xy - 4x - 6y = 0$, and adding 24 to both sides gives $(x-6)(y-4) = 24$. Since $x, y \ge 1$, we have $x - 6 \ge -5$ and $y - 4 \ge -3$. Neither factor can be zero. If both were negative, their product would be at most $5 \cdot 3 = 15 < 24$, so both are positive. Then each positive divisor d of 24 gives exactly one solution, $x = 6 + d$ and $y = 4 + 24/d$. Since $24 = 2^3 \cdot 3$ has $4 \cdot 2 = 8$ positive divisors, there are 8 ordered pairs.

Answer: 8

Problem 2

How many pairs of positive integers (x, y) satisfy $x^2 = y^2 + 6y + 129$?

Since $y^2 + 6y + 129 = (y+3)^2 + 120$, the equation is $x^2 - (y+3)^2 = 120$, that is, $(x - y - 3)(x + y + 3) = 120$. The second factor is positive, so the first is positive too. The two factors differ by $2y + 6$, which is even, so they have the same parity, and as their product 120 is even they are both even. Also the second factor is larger, by $2y + 6 \ge 8$.

Writing $120 = 2u \cdot 2v$ with $uv = 30$ and u smaller than v, the possibilities are $(2u, 2v) = (2, 60)$, $(4, 30)$, $(6, 20)$ and $(10, 12)$. Then $x = u + v$ and $y + 3 = v - u$, giving $(x, y) = (31, 26)$, $(17, 10)$, $(13, 4)$ and $(11, -2)$. The last has y negative (its factors differ by only 2), so there are 3 pairs.

Answer: 3

Problem 3

A post office sells only 7p and 12p stamps. In how many different ways can a parcel be given exactly £5 of postage? (Two ways are the same if they use the same number of each stamp.)

We need non-negative integers with $7x + 12y = 500$. Modulo 7, we have $12 \equiv 5$ and $500 = 71 \cdot 7 + 3$, so $5y \equiv 3 \pmod 7$. The inverse of 5 modulo 7 is 3, since $15 \equiv 1$, so $y \equiv 9 \equiv 2 \pmod 7$.

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Conversely, any $y \equiv 2 \pmod 7$ with $12y \leq 500$ makes $500 - 12y$ a non-negative multiple of 7. The condition $12y \leq 500$ means $y \leq 41$, and the values $y \equiv 2 \pmod 7$ in that range are $y = 2, 9, 16, 23, 30, 37$, giving $x = 68, 56, 44, 32, 20, 8$. So there are 6 ways.

Answer: 6

Problem 4

Find all positive integers n for which $n! + 5n + 13$ is a perfect square.

For $n \geq 5$, both $n!$ and $5n$ are multiples of 5, so $n! + 5n + 13 \equiv 13 \equiv 3 \pmod 5$. But the squares modulo 5 are $0^2 \equiv 0$, $(\pm 1)^2 \equiv 1$ and $(\pm 2)^2 \equiv 4$, so a square is never $\equiv 3 \pmod 5$. Hence $n \leq 4$, and we check the four cases. Here $n = 1$ gives 19, $n = 2$ gives $25 = 5^2$, $n = 3$ gives 34 and $n = 4$ gives 57. The only answer is $n = 2$.

Answer: n = 2

Problem 5

Find all pairs of positive integers (x, y) such that $x^3 - y^3 = 721$.

Write $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. The second factor is positive, so $d = x - y$ is a positive divisor of $721 = 7 \cdot 103$. Since $x^2 + xy + y^2 > x^2 - 2xy + y^2 = d^2$, we get $721 = d(x^2 + xy + y^2) > d^3$, so $d \leq 8$, and therefore $d = 1$ or $d = 7$.

Substitute $x = y + d$, so that $x^2 + xy + y^2 = 3y^2 + 3dy + d^2$. If $d = 1$, then $3y^2 + 3y + 1 = 721$, so $y^2 + y = 240 = 15 \cdot 16$, giving $y = 15$ and $x = 16$. If $d = 7$, then $3y^2 + 21y + 49 = 103$, so $y^2 + 7y - 18 = 0$, that is, $(y + 9)(y - 2) = 0$, giving $y = 2$ and $x = 9$. We check that $16^3 - 15^3 = 4096 - 3375 = 721$ and $9^3 - 2^3 = 729 - 8 = 721$. The solutions are $(9, 2)$ and $(16, 15)$.

Answer: (9, 2) and (16, 15)

Problem 6

How many ordered triples of positive integers (a, b, c) satisfy $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{7}{10}$?

First find the triples with $a \leq b \leq c$. Since $\frac{1}{a}$ is the largest of the three fractions, $\frac{3}{a} \geq \frac{7}{10}$, so $a \leq 4$, and $\frac{1}{a} < \frac{7}{10}$, so $a \geq 2$. In each case, $\frac{1}{b}$ is the larger of the two remaining fractions, so $\frac{2}{b}$ is at least what remains.

If $a = 2$, then $\frac{1}{b} + \frac{1}{c} = \frac{1}{5}$, so $5 < b \leq 10$. Here $\frac{1}{c} = \frac{b-5}{5b}$, which is a unit fraction for $b = 6$ (giving $c = 30$) and $b = 10$ (giving $c = 10$), but not for $b = 7, 8, 9$. If $a = 3$, then $\frac{1}{b} + \frac{1}{c} = \frac{11}{30}$, so $3 \leq b \leq \frac{60}{11}$, that is, $b \in \{3, 4, 5\}$. Then $\frac{1}{c} = \frac{11}{30} - \frac{1}{b}$ equals $\frac{1}{30}$, $\frac{7}{60}$ and $\frac{1}{6}$ respectively, so $b = 3, c = 30$ and $b = 5, c = 6$ work. If $a = 4$, then $\frac{1}{b} + \frac{1}{c} = \frac{9}{20}$, so $4 \leq b \leq \frac{40}{9}$, forcing $b = 4$ and $\frac{1}{c} = \frac{1}{5}$, so $c = 5$.

So the triples with $a \leq b \leq c$ are $(2, 6, 30)$, $(2, 10, 10)$, $(3, 3, 30)$, $(3, 5, 6)$ and $(4, 4, 5)$. Those with three different entries give 6 ordered triples each, and those with a repeated entry give 3 each. The total is $6 + 3 + 3 + 6 + 3 = 21$.

Answer: 21